



NELIREF

New Life Research
Foundation (NELIREF) Ltd/Gte

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3rd NELIREF Data Science & AI Summer School

*Roadmap to Data Science & Artificial Intelligence
and Pre-Lab*

Instructor: Joseph Luper Tsenum

June 19-27, 2026



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Why Artificial Intelligence as a Christian?

Jeremiah 1:9-10

Then the LORD put forth His hand and touched my mouth, and the LORD said to me: "Behold, I have put My words in your mouth. See, I have this day set you over the nations and over the kingdoms, To root out and to pull down, To destroy and to throw down, To build and to plant."

Jeremiah 51:20-23

"You are My battle-ax and weapons of war: For with you I will break the nation in pieces; With you I will destroy kingdoms; With you I will break in pieces the horse and its rider, the chariot and its rider, man and woman, old and young, the young man and the maiden, the shepherd and his flock, the farmer and his yoke of oxen, and governors and rulers."

Psalms 2:1-3

Why do the nations rage, and the people plot a vain thing? The kings of the earth set themselves, and the rulers take counsel together, against the LORD and against His Anointed, saying, "Let us break Their bonds in pieces and cast away Their cords from us."

What is Artificial Intelligence?



“AI is a machine’s ability to perform the cognitive functions we associate with human minds, such as perceiving, reasoning, learning, interacting with an environment, problem solving, and even exercising creativity.”

— McKinsey

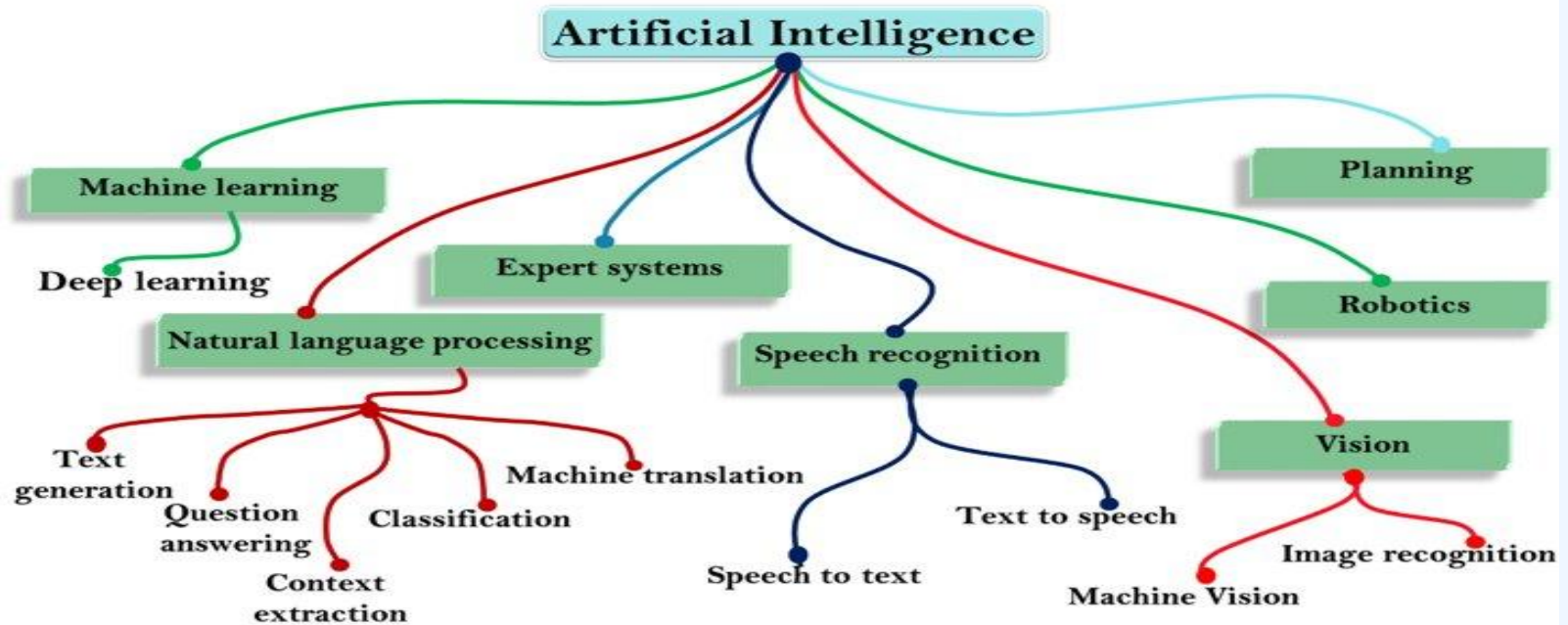


You interact with AI daily without realizing it such as Siri voice assistants, customer service chatbots, Generative AI such as ChatGPT, gaming, predictive models for risk/protective factors in healthcare, forecasting models in businesses, mobile banking, fraud detection, customer engagement, social media such as Facebook, etc.



Driverless cars, scheduling of elevators, self-navigating vacuum cleaners are some of the applications of Reinforcement Learning.

Branches of Artificial Intelligence



Source: tekrajawasthi15.medium.com — The Complete Roadmap to be a Data Scientist

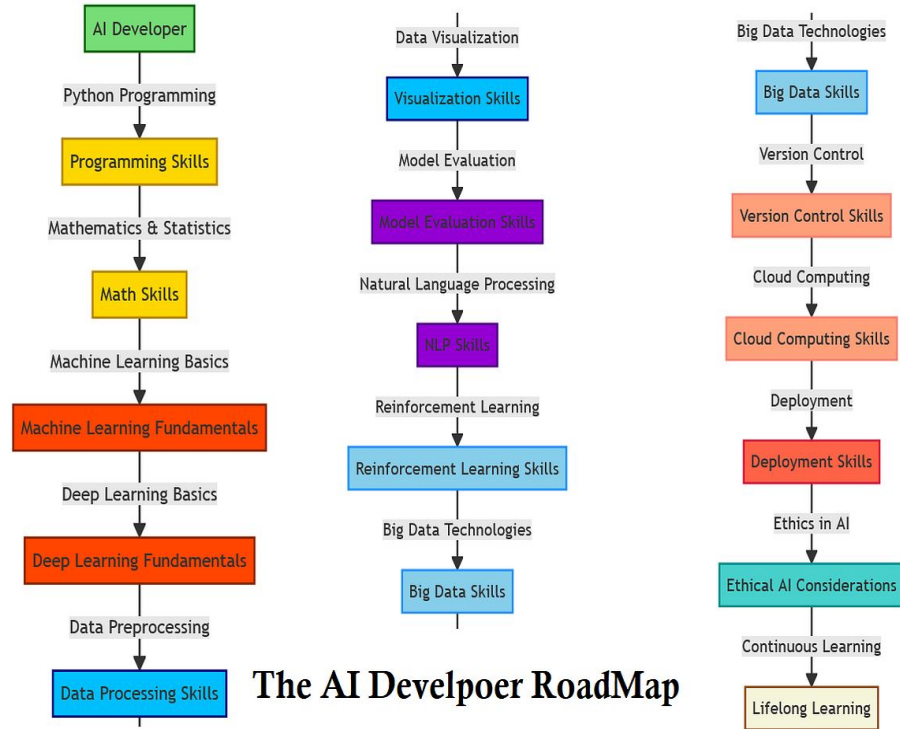
Roadmap to Artificial Intelligence & Machine Learning



AI/ML study resources compiled by New Life Research Foundation Training Department. Check our platforms for details.



Apply for Financial Aid on Coursera and take free online courses such as Math of Machine Learning by DeepLearning.AI (Andrew Ng). Check our platforms for details.



The AI Developer RoadMap

Hard Skills Needed in the Era of AI/ML



Possess hard skills that will be helpful in our pioneering work and flagship projects.



Bash scripting, ability to work in a Linux/Unix environment



Programming skills: Python
Others: R, SQL, Java, C & C++, MATLAB, Perl, Ruby, etc.



Machine learning (supervised & unsupervised),
generative/automated models



Interpretable AI such as LIME, SHAP, Anchors



Artificial intelligence and machine learning are implemented using
Python programming language (and others such as R)



Database management; statistical analysis; mobile development;
network security



PowerBI, Tableau, ability to use LaTeX editor, AI assistants, vibe
coding



Adobe software suite such as Adobe Illustrator, etc.

Soft Skills Needed in the Era of AI/ML



Hard skills are not sufficient for research in a tech century — every researcher is expected to also possess soft skills.

Integrity | Discipline

Emotional Intelligence (empathy)

Dependability

Effective Communication

Presentation Skills

Open-Mindedness

Teamwork

Creativity

Problem-Solving

Critical Thinking

Adaptability

Organization

Willingness to Learn

Leadership

Time Management

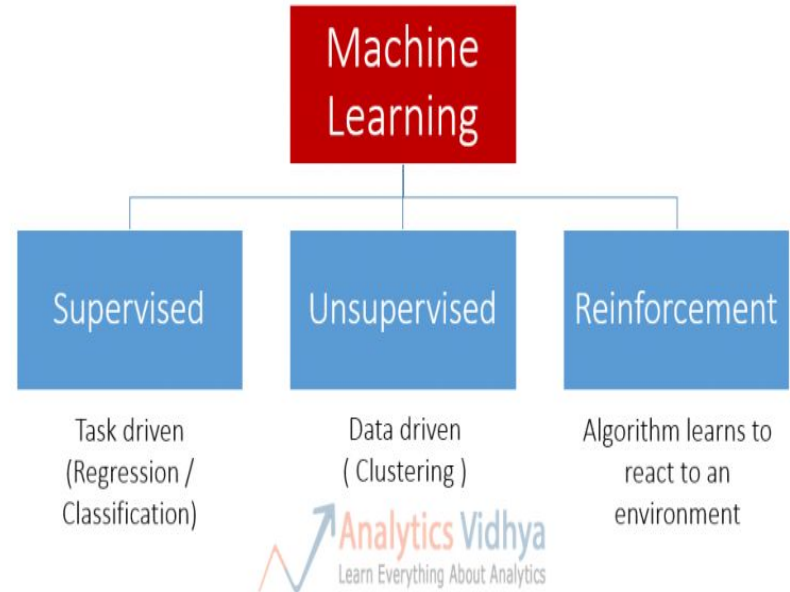
Work Ethic

Conflict Management

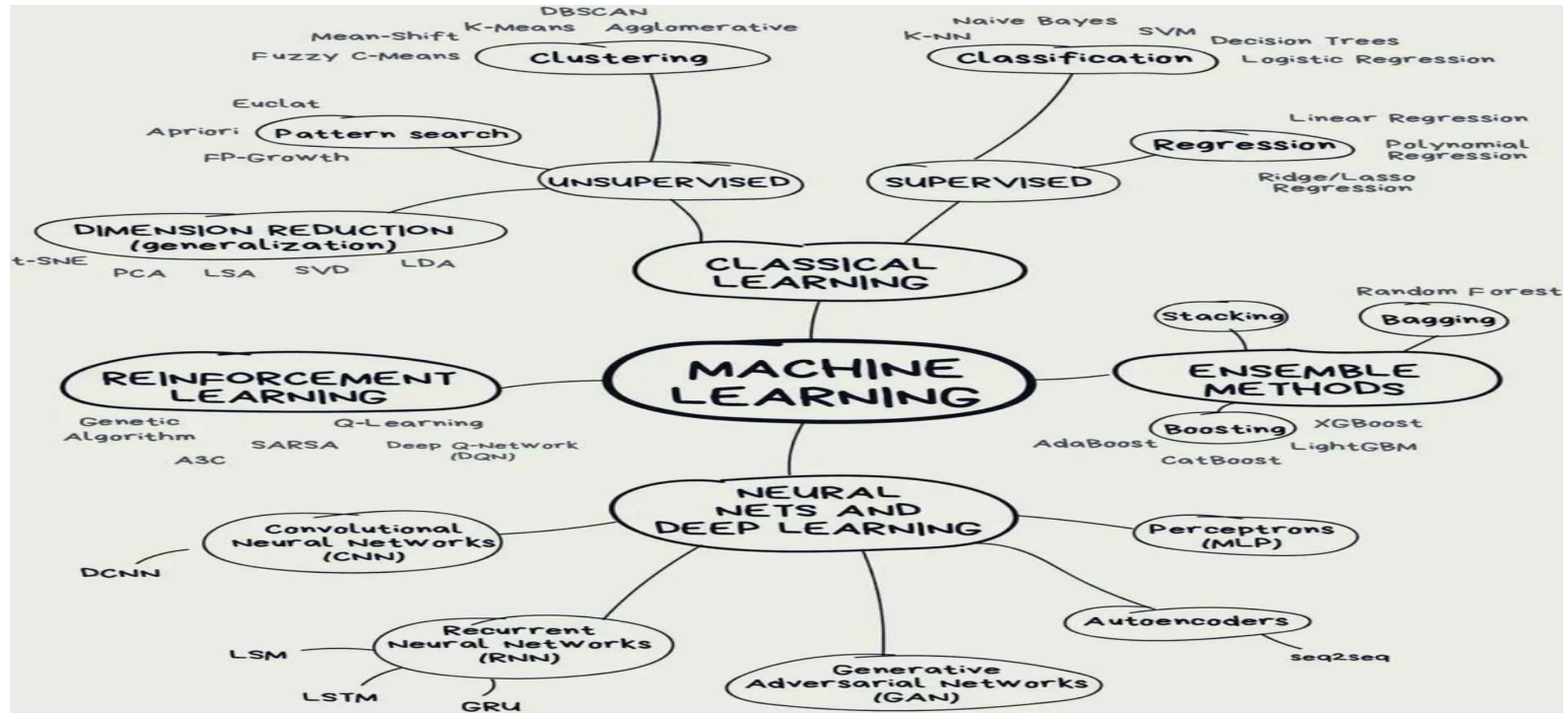
What is Machine Learning?

- Machine Learning is the science of teaching machines how to learn by themselves. It is a type of artificial intelligence based on algorithms that are trained on data.
- In the 1970s, machine learning was applied in medical-imaging analysis and high-resolution weather forecasting.
- Today, ML can learn from data and detect patterns thereby making predictions to help in decision and policy making.
- Deep Learning is a type of machine learning that can process large amounts of data such as images, bulk biological data, etc.
- DL requires less human intervention and can produce more accurate results (unsupervised learning) than traditional or classical machine learning methods (supervised learning). New patterns are learned.

Types of Machine Learning

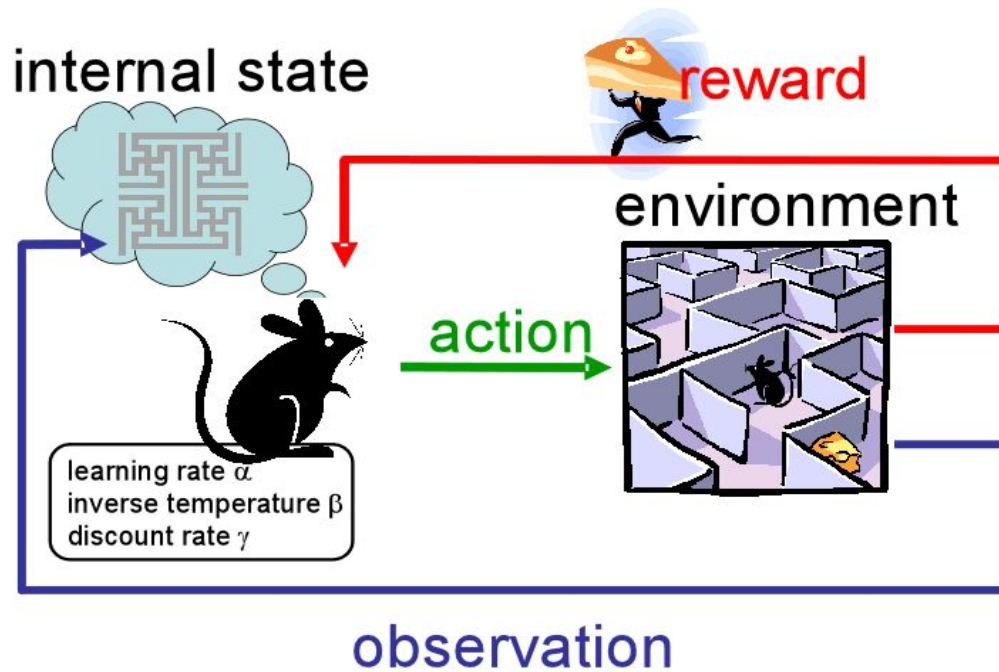


Machine Learning Methods Overview



Source: Atul Kumar

Reinforcement Learning



Source: UTCS RL Reading Group

The pet

is the artificial agent

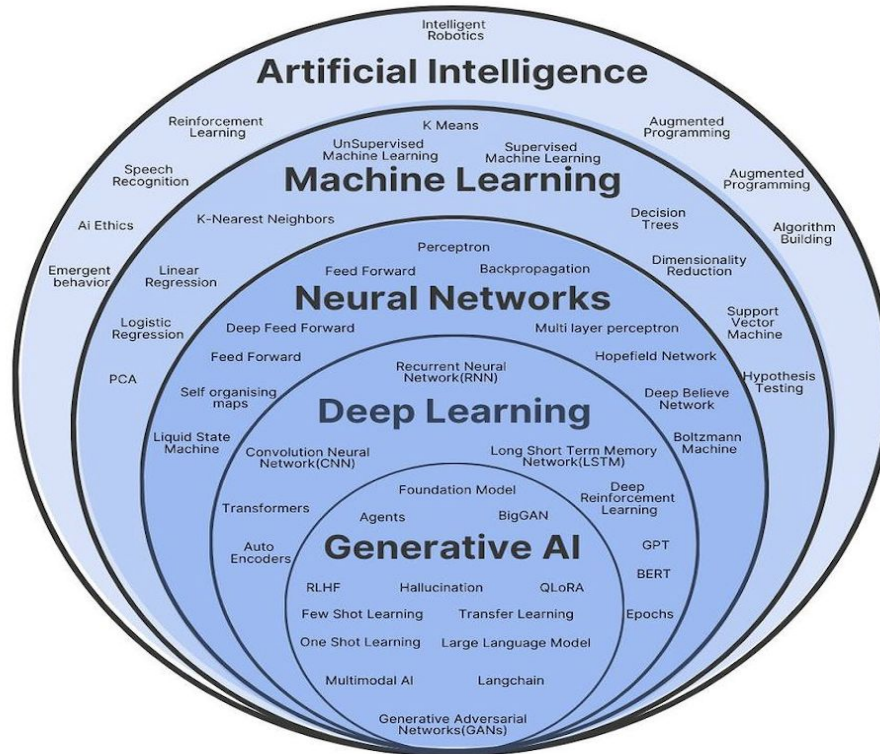
The good behavior

is the resultant action

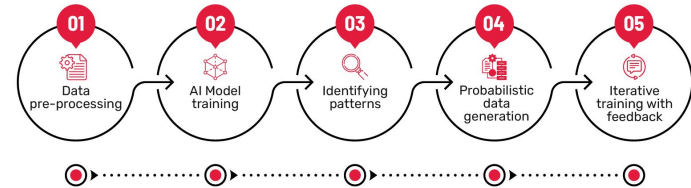
The treatment

is the reward function

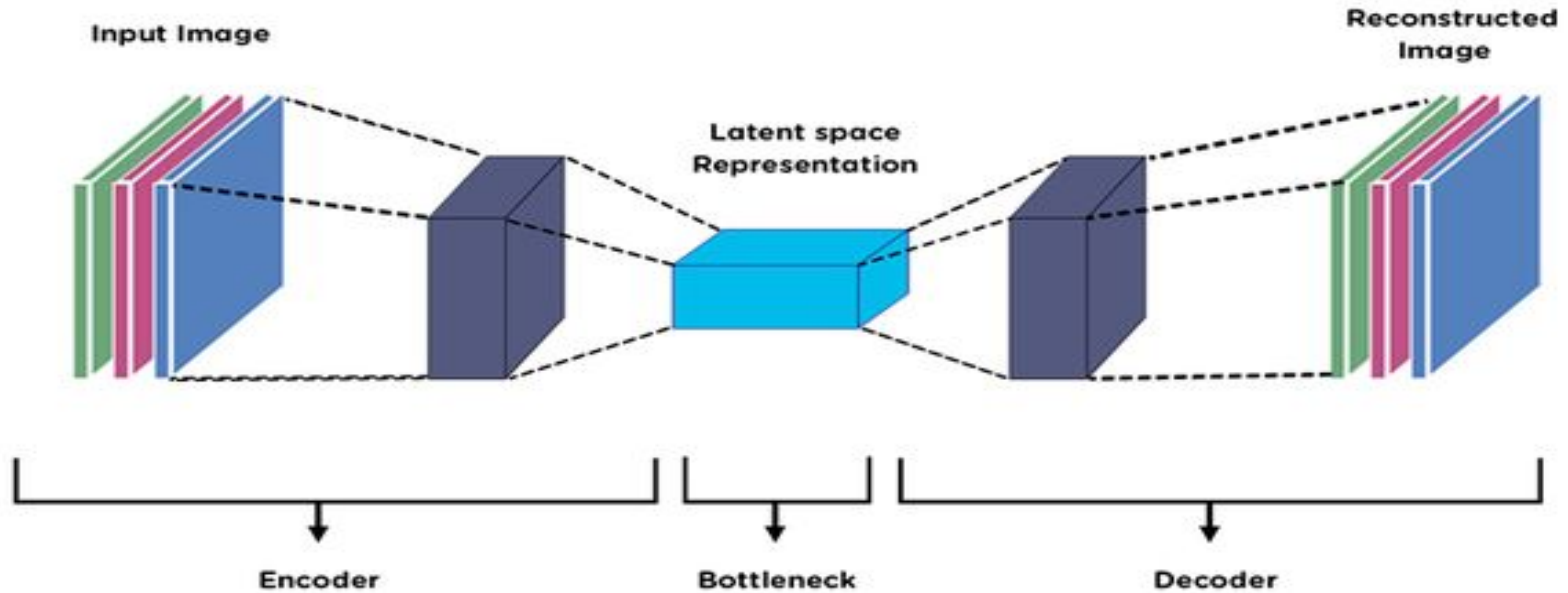
AI Revolution



How Generative AI Works



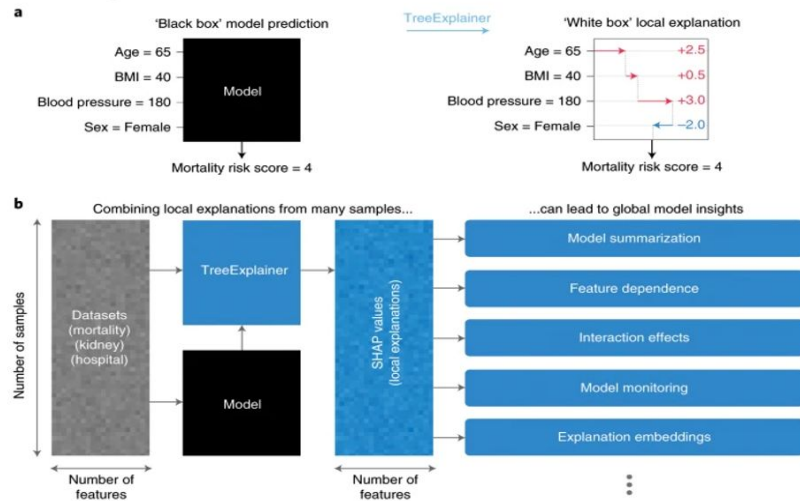
Autoencoders



Source: Premanand S (Analytics Vidhya)

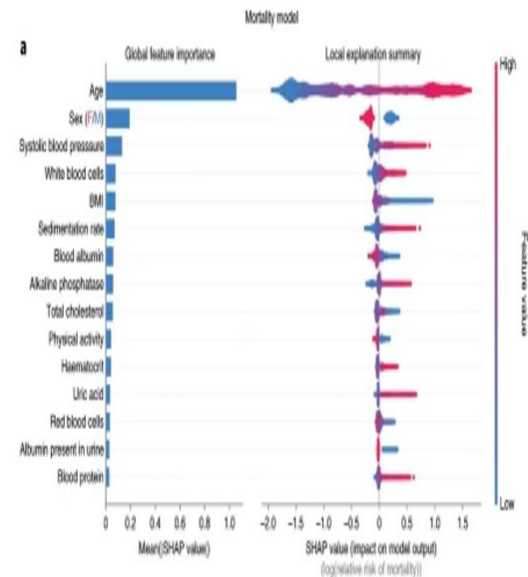
Increasing Demand for Explainable Black-Box Models

Fig. 1: Local explanations based on TreeExplainer enable a wide variety of new ways to understand global model structure.



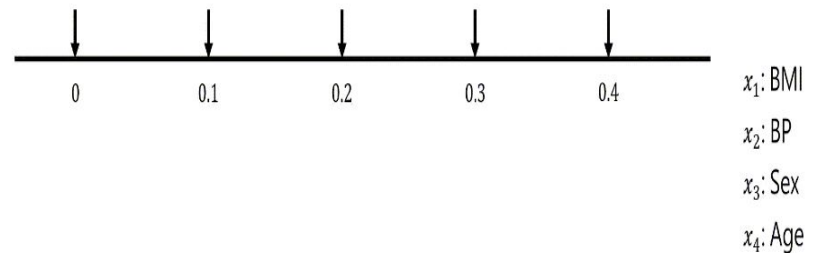
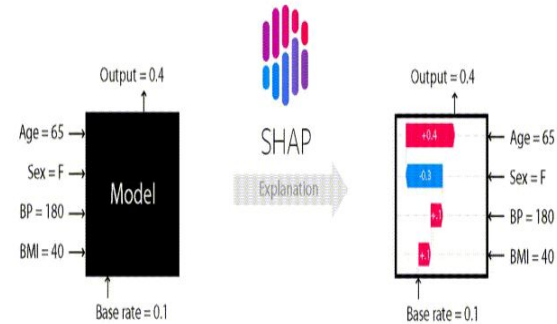
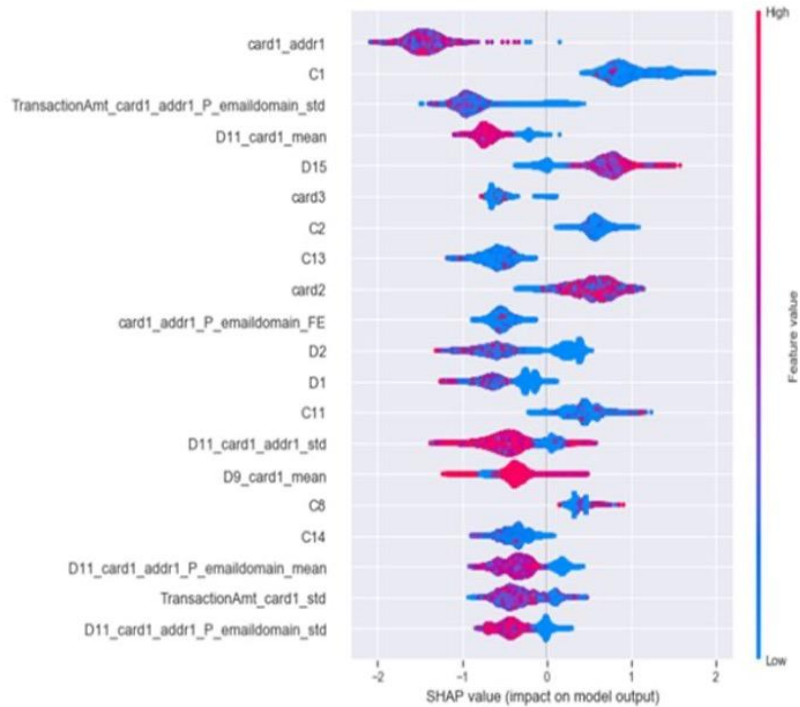
a. A local explanation based on assigning a numeric measure of credit to each input feature. **b.** By combining many local explanations, we can represent global structure while retaining local faithfulness to the original model. We demonstrate this by using three medical datasets to train gradient boosted decision trees and then compute local explanations based on SHAP values². Computing local explanations across all samples in a dataset enables development of many tools for understanding global model structure.

Fig. 4: By combining many local explanations, we can provide rich summaries of both an entire model and individual features.



Explainable Ensemble & Deep Learning

Using SHAP, LIME, and Anchors



Courtesy: Shilpi Bhattacharyya

Pre-Lab

Core Tools (IDEs — Integrated Development Environments) & Libraries by Topic

Core Tools

- Anaconda or Miniconda (for managing environments and packages)
- Jupyter Notebook / JupyterLab · Google Colab or Anaconda (optional browser-based platform)
- Python (version 3.9 or 3.10 and above)

Statistics for Machine Learning

numpy, scipy, pandas, matplotlib, seaborn, statsmodels

Object-Oriented Programming in Python

No external libraries (uses built-in dataclasses)

Python Programming (Intro & Intermediate)

numpy, pandas, matplotlib, plus built-in random, math, itertools

Supervised ML & Deep Learning

scikit-learn, xgboost, lightgbm, tensorflow or pytorch, keras

NLP, LLMs/Foundation Models & Generative AI

PyTorch, scikit-learn, transformers (Hugging Face)

Pre-Lab — Installation Guide



macOS

Core Tools:

- Anaconda or Miniconda
- Jupyter Notebook / JupyterLab
- Google Colab (optional)
- Python (3.9 or 3.10)

Libraries by Topic:

- Statistics: numpy, scipy, pandas, matplotlib, seaborn, statsmodels
- Python Programming (Intro & Intermediate), plus built-in random



Ubuntu Linux

Core Tools:

- Anaconda or Miniconda
- Jupyter Notebook / JupyterLab
- Google Colab (optional)

Libraries by Topic:

- Statistics: numpy, scipy, pandas, matplotlib, seaborn, statsmodels
- OOP in Python: no external libraries (built-in dataclasses)
- Python Programming (Intro & Intermediate)



Windows

Core Tools:

- Anaconda or Miniconda
- Jupyter Notebook / JupyterLab
- Google Colab (optional)
- Python (3.9 or 3.10)

Libraries by Topic:

- Supervised ML & Deep Learning: scikit-learn, xgboost, lightgbm, tensorflow/pytorch, keras
- PyTorch, scikit-learn, transformers (Hugging Face)

IDEs — Integrated Development Environments

Terminal Commands and browser-based platform



Miniconda and Jupyter Notebook Installation (Windows, Ubuntu Linux, macOS)



- Leaving PowerShell and opening Command Prompt (`cmd`) for Window users
- Downloading Miniconda at
- Installing Miniconda
- Adding Miniconda to PATH
- Creating a conda environment
- Installing Jupyter Notebook/JupyterLab
- Launching Jupyter, running and saving cells



Google Colab and Google Drive / Jupyter Notebook



- Signing in with Gmail on Google Colab or Google Drive
<https://colab.research.google.com/> | <https://drive.google.com/>
- Uploading `.ipynb` notebooks
- Uploading datasets
- Opening notebooks from Google Drive
- Using Google Colab directly from Drive
- Running and saving notebooks in the browser (automatically saves progress)



Using Jupyter Notebook Online via Anaconda

- Signing up on the Anaconda website
<https://anaconda.org/channels/anaconda/packages/python/overview>
- Navigating from the Python overview page
- Opening Jupyter Notebook/JupyterLab online
- Uploading notebooks and datasets
- Running and saving notebooks in the browser

Pre-Lab — Installation Guide

Core Tools & Libraries — Terminal Commands



macOS

- Install Homebrew: `/bin/bash -c "$(curl -fsSL raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`
- Install Python: `brew install python@3.10`
- Install Miniconda: docs.conda.io/en/latest/miniconda.html | <https://www.anaconda.com/docs/getting-started/miniconda/install/overview>
- Install Jupyter: `conda install jupyterlab`
- Install libraries: `pip install numpy scipy pandas matplotlib seaborn statsmodels scikit-learn xgboost lightgbm tensorflow pytorch keras transformers` | use VS Code (optional)



Ubuntu Linux

- Update: `sudo apt update` & `sudo apt upgrade`
- Install Python: `sudo apt install python3 python3-pip`
- Download Miniconda and install: `bash Miniconda3-latest-Linux-x86_64.sh` | <https://www.anaconda.com/docs/getting-started/miniconda/install/overview>
- Install Jupyter: `pip install jupyterlab`
- Install libraries: same as above via `pip` or `conda`



Windows

- Download Miniconda: docs.conda.io/en/latest/miniconda.html | <https://www.anaconda.com/docs/getting-started/miniconda/install/overview>
- Use Anaconda Prompt to create env: `conda create -n dsaipy python=3.10`
- Activate: `conda activate dsaipy`
- Install Jupyter: `conda install jupyterlab`
- Install libraries: `pip install the listed packages`



Announcement

NELIREF 4th Annual Conference comes up in August 2026 — visit our website and social media platforms for more details

<https://neliref.org/>

Q / A

References

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Thank You!



**We admonish you to live
for God and connect with
our community!**



Connect with Us!



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